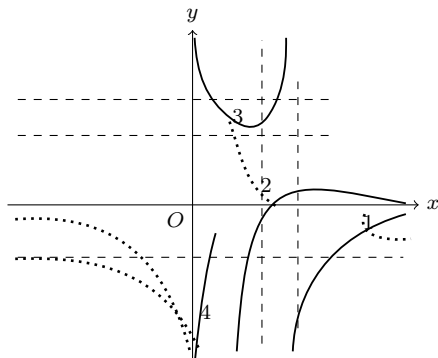


lying wholly between the two asymptotes $y = 0$, $y = -1$; the latter one of the branches belonging to the cuspidal asymptote $x = 0$; this last branch cuts the asymptote $x = 0$ at $y = -2$, and then, cutting the asymptote $y = -1$ and $x = -\frac{2}{7}$ ($= -0 \cdot 3$), goes on to touch at infinity the asymptote $y = 0$. It is now easy to trace the curve.

The figure shows the two curves. The curve (1) is shown by a continuous line, the curve (2) by a thick dotted line; the points 1, 2, 3, 4 show the above mentioned



four intersections of the curves; the point 1 and the dotted branch through it are of necessity drawn considerably out of their true positions; viz. as above appearing, the x -coordinate of 1 is $= 17 \cdot 00$, and the equation of the vertical tangent to the branch is $x = 10 \cdot 8$.

[Vol. xxii., pp. 78, 79.]

4620. (Proposed by A. B. Evans, M.A.)—Find the least integral values of x and y that will satisfy the equation

$$x^2 - 953y^2 = -1.$$

Solution by Professor Cayley.

The values are given in Degen's Tables, viz.

$$x = 2746864744, \quad y = 88979677.$$

The work referred to is entitled "Canon Pellianus, sive Tabula simplicissimam æquationis celeberrimæ $y^2 = ax^2 + 1$ solutionem pro singulis numeri dati valoribus ab 1 usque ad 1000 in numeris rationalibus iisdemque integris exhibens. Auctore C. F. Degen, Hafniæ (Copenhagen), 1817."

Table I., pp. 3–106 gives, for all numbers 1 to 1000, the denominators, (?) the quotients of the convergent fraction of $\sqrt{a}$, and also the least values of x, y which will satisfy the equation $x^2 - ay^2 = +1$. Thus

953	30,	1,	6,	1,	2,	1,	3,	8,	1,	1,	(4,	4),
	1,	53,	8,	41,	17,	37,	16,	7,	32,	29,	(13,	13),
	488830275367615376, 15090531843660371073.											

Table II., pp. 109–112, is described as giving for all those values of a between 1 and 1000, for which there exists a solution of the equation $x^2 - ay^2 = -1$, the least values of x and y which satisfy this equation: thus 953, x and y as above. It is, however, to be noticed that the values of $a = \beta^2 + 1$, for which there is the obvious solution $x = \beta$, $y = 1$, are omitted from the table. The reason for this appears, but the heading should have been different.

[Vol. xxiii., January to July, 1875, pp. 18, 19.]

4528. (Proposed by Professor Cayley.)—A lottery is arranged as follows:—There are n tickets representing a, b, c pounds respectively. A person draws once; looks at his ticket; and if he pleases, draws again (out of the remaining $n - 1$ tickets); looks at his ticket, and if he pleases draws again (out of the remaining $n - 2$ tickets); and so on, drawing in all not more than k times; and he receives the value of the last drawn ticket. Supposing that he regulates his drawings in the manner most advantageous to him according to the theory of probabilities, what is the value of his expectation?

Solution by the Proposer.

Let the expression “ a or α ” signify “ a or α , whichever of the two is greatest,” and let $M_1(a, b, c, \dots)$ denote the mean of the quantities $(a, b, c, \dots)$, viz. their sum, divided by the number of them.

To fix the ideas, consider five quantities a, b, c, d, e , and write

$$M_1(a, b, c, d, e) = M_1(a, b, c, d, e),$$

$$M_2(a, b, c, d, e) = M_1\{a \text{ or } M_1(b, c, d, e), b \text{ or } M_1(a, c, d, e), \dots, e \text{ or } M_1(a, b, c, d)\},$$

$$M_3(a, b, c, d, e) = M_1\{a \text{ or } M_2(b, c, d, e), b \text{ or } M_2(a, c, d, e), \dots, e \text{ or } M_2(a, b, c, d)\},$$

and so on. And the like in the case of any number of quantities $a, b, c, \dots$

Then the value of the expectation is $= M_k(a, b, c, \dots)$.

For, when $k = 1$, the value is obviously $= M_1(a, b, c, \dots)$.

When $k = 2$, if a is drawn, the adventurer will be satisfied or he will draw again, according as a or $M_1(b, c, \dots)$ is greatest, viz. in this case the value of the expectation is " a or $M_1(b, c, \dots)$."

So if b is drawn, the adventurer will be satisfied or he will draw again, according as b or $M_1(a, c, \dots)$ is greatest; viz. in this case the value of the expectation is " b or $M_1(a, c, \dots)$ "; and so on: and the several cases being equally probable, the value of the total expectation is

$$= M_1\{a \text{ or } M_1(b, c, \dots), b \text{ or } M_1(a, c, \dots), \dots\} = M_2(a, b, c, \dots) :$$

and the like for $k = 3, k = 4$, &c.

For instance, $a, b, c, d = 1, 2, 3, 4$, $M_1(1, 2, 3, 4) = \frac{10}{4}$,

$$M_2(1, 2, 3, 4) = M_1(1 \text{ or } \frac{9}{3}, 2 \text{ or } \frac{8}{3}, 3 \text{ or } \frac{7}{3}, 4 \text{ or } \frac{6}{3}) = M_1(\frac{9}{3}, \frac{8}{3}, \frac{3}{1}, \frac{12}{3}) = \frac{38}{12},$$

$$M_2(2, 3, 4) = M_1(2 \text{ or } \frac{7}{2}, 3 \text{ or } \frac{6}{2}, 4 \text{ or } \frac{5}{2}) = M_1(\frac{7}{2}, \frac{6}{2}, \frac{8}{2}) = \frac{21}{6},$$

$$M_2(1, 3, 4) = M_1(1 \text{ or } \frac{7}{2}, 3 \text{ or } \frac{5}{2}, 4 \text{ or } \frac{4}{2}) = M_1(\frac{7}{2}, \frac{6}{2}, \frac{8}{2}) = \frac{21}{6},$$

$$M_2(1, 2, 4) = M_1(1 \text{ or } \frac{6}{2}, 2 \text{ or } \frac{5}{2}, 4 \text{ or } \frac{3}{2}) = M_1(\frac{6}{2}, \frac{5}{2}, \frac{8}{2}) = \frac{19}{6},$$

$$M_2(1, 2, 3) = M_1(1 \text{ or } \frac{5}{2}, 2 \text{ or } \frac{4}{2}, 3 \text{ or } \frac{3}{2}) = M_1(\frac{5}{2}, \frac{4}{2}, \frac{6}{2}) = \frac{15}{6},$$

$$M_3(1, 2, 3, 4) = M_1(1 \text{ or } \frac{21}{6}, 2 \text{ or } \frac{21}{6}, 3 \text{ or } \frac{19}{6}, 4 \text{ or } \frac{15}{6})$$

$$= M_1(\frac{21}{6}, \frac{21}{6}, \frac{19}{6}, \frac{24}{6}) = \frac{85}{24},$$

$$M_3(1, 2, 3) = 3 \text{ \&c.},$$

$$M_4(1, 2, 3, 4) = M_1(1 \text{ or } 4, 2 \text{ or } 4, 3 \text{ or } 4, 4 \text{ or } 3) = M_1(4, 4, 4, 4) = 4.$$

Or finally

$$M_1, M_2, M_3, M_4 = \frac{10}{4}, \frac{38}{12}, \frac{85}{24}, 4 = \frac{60}{24}, \frac{76}{24}, \frac{85}{24}, \frac{96}{24}.$$

Cor. If the $a, b, c, \dots$ denote penalties instead of prizes, then the solution is the same, except that " a or α " must now denote " a or α , whichever of them is least."

[Vol. xxiii., pp. 47, 48.]

4581. (Proposed by the Rev. M. M. U. Wilkinson.)—A witness, whose statement is what he opines once in m times, and whose opinion is correct once in n times, asserted that the number of a note, issued by a bank universally known to have issued notes numbered from B to $B + A - 1$ inclusive, was $B + P$, where P is either $0, 1, 2, \dots$, or $A - 1$. Prove (1) that the probability that the note in question was that note is

$$\frac{1}{mn} \left\{ 1 + \frac{(m-1)(n-1)}{A-1} \right\}.$$

The above witness also said that the note was signed by X , it being universally known that X has signed one note, and Y the remaining $A - 1$ notes; find (2) the probability that this last statement was correct.

Remark by Professor Cayley.

There is a serious difficulty in the question, or the answer; I think, in the question. Try the answer in numbers $m = 10$, $n = 10$. The witness says what he opines once out of 10 times—he is in fact an atrocious liar; and he opines rightly once out of 10 times, that is, wrongly 9 times out of 10; he is therefore a blunderer—but a remarkably ingenious one, in that the chances are so greatly against his blundering upon a right result.

He says that the note was signed by X , and the chance of this being so is found to be $\frac{1}{100} + \frac{81}{100} = \frac{82}{100}$, or more than $\frac{8}{10}$; the larger part $\frac{81}{100}$ of this is obtained as follows:—the witness having said that the note was signed by X , the chances are 9 out of 10 that he thought the reverse; and, thinking the reverse, the chance is 9 out of 10 that he thought wrongly, viz. that the note was signed by X . But can the statement of such a witness create any probability in favour of the event?

The fallacy seems to consist in the assumption that n can have a determinate value irrespective of the nature of the opinion. Suppose there are 500 notes, and that the opinion is that the note was a definite number 99; it is quite conceivable that, in forming a series of such opinions, the witness may be wrong 9 times out of 10. But let the opinion be that the note was *not* 99; no amount of ingenuity of blundering can make him wrong 9 times out of 10 in a series of such opinions. If it could, a friend who knew the true opinion of the witness, would be able 9 times out of 10 to know the number of the note, from the mere fact that the witness opines that the note is not a named number.

[Vol. xxiii., p. 58.]

4638. (Proposed by Professor Cayley.)—Find the equation of the surface which is the envelope of the quadric surface

$$ax^2 + by^2 + cz^2 + dw^2 = 0,$$

where a, b, c, d are variable parameters connected by the equation

$$Abc + Bca + Cab + Fad + Gbd + Hcd = 0;$$

and consider in particular the case in which the constants A, B, C, F, G, H satisfy the condition

$$(AF)^{\frac{1}{2}} + (BG)^{\frac{1}{2}} + (CH)^{\frac{1}{2}} = 0.$$

[Vol. xxiv., July to December, 1875, p. 41.]

4694. (Proposed by Professor Cayley.)—Taking F, F' a pair of reciprocal points in respect to a circle, centre O ; then if F, F' are centres of force, each force varying as (distance) $^{-n}$, prove that (1) the resultant force upon any point P on the circle is in the direction of a fixed point S on the axis OFF' ; and if, moreover, the forces at the unit of distance are as $(OF)^{\frac{1}{2}(n-1)}$ to $(OF')^{\frac{1}{2}(n-1)}$, then (2) the resultant force is proportional to

$$(SP)^{-\frac{1}{2}(n-1)} \cdot (PV)^{-\frac{1}{2}(n+1)},$$

where PV is the chord through S .

[Vol. xxiv., pp. 72-74.]

4793. (Proposed by Professor Wolstenholme, M.A.)—If $y = x^n(\log x)^r$, where n and r are integers, prove that

$$x^r \frac{d^{n+r}y}{dx^{n+r}} + \frac{r(r-1)}{2} x^{r-1} \frac{d^{n+r-1}y}{dx^{n+r-1}} + \frac{r(r-1)(r-2)(3r-5)}{24} x^{r-2} \frac{d^{n+r-2}y}{dx^{n+r-2}} + \dots \\ \dots + (2^{r-1} - 1)x^2 \frac{d^{n+2}y}{dx^{n+2}} + x \frac{d^{n+1}y}{dx^{n+1}} = [r]^r [n]^n,$$

the coefficients being

$$\frac{\Delta^{r-1} 1^{r-1}}{[r-1]}, \quad \frac{\Delta^{r-2} 1^{r-1}}{[r-2]}, \quad \frac{\Delta^{r-3} 1^{r-1}}{[r-3]}, \quad \dots, \quad \frac{\Delta 1^{r-1}}{[1]}, \quad \text{and } 1;$$

so that the result may be symbolically written

$$e^{xD\Delta} \left(1^{r-1} \frac{d^{n+1}y}{dx^{n+1}} \right) = \frac{[r]^r [n]^n}{x},$$

where D denotes $\frac{d}{dx}$ and operates on $\frac{d^n y}{dx^n}$ only, and Δ operates on 1^{r-1} only, the terms after the r th all vanishing since $\Delta^m x^n = 0$, when m is an integer $> n$. The calculations involved prove that, when $x = 1$,

$$\Delta^{n-1} x^n = \frac{n+1}{2}, \quad \Delta^{n-2} x^n = [n+1] \frac{3n-2}{24}, \\ \Delta^{n-3} x^n = [n+1] \frac{(n-1)(n-2)}{48}.$$

Solution by Professor Cayley.

Since $y = x^n(\log x)^r$, therefore $(xd_x - n)y = rx^n(\log x)^{r-1}$; by repeating the same operation, we have

$$(xd_x - n)^r y = [r]^r x^n; \quad \text{whence } d_x^n (xd_x - n)^r y = [r]^r [n]^n.$$

Now, for any value whatever of the function y , we have

$$d_x^n (xd_x - n)^r y = Ax^r d_x^{r+n} y + Bx^{r-1} d_x^{r+n-1} y + Cx^{r-2} d_x^{r+n-2} y + \&c.,$$

the coefficients $A, B, C, \dots$ being functions, presumably of r, n , but independent of the form of the function y . It will, however, appear that $A, B, C, \dots$ are, in fact, functions of r only.

To see how this is, observe that $(xd_x - n)^r$ consists of a set of terms

$$(xd_x)^\theta, \quad (\theta = 0 \text{ to } r),$$

where $(xd_x)^\theta$ denotes θ repetitions of the operation xd_x ; by a well-known theorem, this is $[xd_x + \theta - 1]^\theta$, where, after expansion of the factorial, $(xd_x)^\theta$ is to be replaced by $x^\theta d_x^\theta$, thus

$$(xd_x)^2 = [xd_x + 1]^2 = x^2 d_x^2 + xd_x, \quad (xd_x)^3 = [xd_x + 2]^3 = x^3 d_x^3 + 3x^2 d_x^2 + 2xd_x, \quad \&c.;$$

thus $(xd_x - n)^r$ consists of a series of terms $x^\theta d_x^\theta$, ($\theta = 0$ to r), and, operating with d_x^n , this last, $(d_x + d_{x'})^n$, consists of a series of terms such as $d_x^\alpha d_{x'}^{n-\alpha}$, where the unaccented symbol operates on the x^θ , and the accented symbol on the y ; the term is thus $x^{\theta-\alpha} d_x^{n+\theta-\alpha}$, or observing that $\theta - \alpha$ is at most $= r$, and putting it $= r - k$, the term is $x^{r-k} d_x^{n+r-k}$, viz. $d_x^n (xd_x - n)^r$ consists of a series of terms of the form $x^{r-k} d_x^{n+r-k}$; or, what is the same thing, $d_x^n (xd_x - n)^r y$ is a series of the form in question.

To understand how it can be that the coefficients $A, B, C, \dots$ are independent of n , take the particular case $r = 2$; then we have here

$$d_x^n (xd_x - n)^2 y = Ax^2 d_x^{n+2} y + Bx d_x^{n+1} y + C d_x^n y.$$

The right-hand side is

$$d_x^n \{x^2 d_x^2 - (2n-1)xd_x + n^2\}y,$$

which is

$$\begin{aligned} &= \{x^2 d_x^{n+2} + 2nxd_x^{n+1} + (n^2 - n)d_x^n\}y \\ &\quad - (2n-1)\{xd_x^{n+1} + nd_x^n\}y + n^2\{d_x^n\}y; \end{aligned}$$

hence

$$A = 1, \quad B = 2n - (2n-1) = 1, \quad C = (n^2 - n) - n(2n-1) + n^2 = 0;$$

and we thus see also how in this particular case the last coefficient is $= 0$, viz. that we have

$$d_x^n (xd_x - n)^2 y = x^2 d_x^{n+2} y + x d_x^{n+1} y,$$

without any term in $d_x^n y$.

To find the coefficients $A, B, C, \dots$ generally, write $y = x^{r+n+\theta}$, then $xd_x - n = r + \theta$, and consequently

$$d_x^n (xd_x - n)^r y = (r + \theta)^r d_x^n x^{r+n+\theta} = (r + \theta)^r [r + n + \theta]^n x^{r+\theta};$$

whence

$$(r + \theta)^r [r + n + \theta]^n = A[r + \theta + n]^{n+r} + B[r + \theta + n]^{n+r-1} + \dots;$$

or, what is the same thing,

$$(r + \theta)^r = A[r + \theta]^r + B[r + \theta]^{r-1} + \dots.$$

Since the left-hand side, and every term $[r + \theta]^s$ on the right-hand side, contains the factor $r + \theta$, there is not on the right-hand side any term $[r + \theta]^0$; dividing the equation by $r + \theta$, it then becomes

$$(r + \theta)^{r-1} = A[r + \theta - 1]^{r-1} + B[r + \theta - 1]^{r-2} + \dots,$$

and we thus have

$$A = \frac{\Delta^{r-1} 1^{r-1}}{[r-1]^{r-1}} (= 1), \quad B = \frac{\Delta^{r-2} 1^{r-1}}{[r-2]^{r-2}};$$

viz. writing $r + \theta = 1 + x$, $u_x = (1 + x)^{r-1}$, and taking the terms in the reverse order, the series is the well-known one

$$u_x = u_0 + \frac{x}{1} \Delta u_0 + \frac{x(x-1)}{1 \cdot 2} \Delta^2 u_0 + \&c.$$

Hence, in general,

$$d_x^n (x d_x - n)^r y = \frac{\Delta^{r-1} 1^{r-1}}{[r-1]^{r-1}} x^r d_x^{r+n} y + \frac{\Delta^{r-2} 1^{r-1}}{[r-2]^{r-2}} x^{r-1} d_x^{r+n-1} y + \&c.,$$

where observe that the last term is $= x d_x^{n+1} y$.

For the function $y = x^n (\log x)^r$, the value of each side is $= [r]^r [n]^n$.

[Vol. xxiv., pp. 89–91.]

4752. (Proposed by Professor Cayley.)—Mr Wolstenholme's Question 3067 may evidently be stated as follows:—

If (a, b, c) are the coordinates of a point on the cubic curve

$$a^3 + b^3 + c^3 = (b+c)(c+a)(a+b),$$

and if

$$(b^2 + c^2 - a^2)x = (c^2 + a^2 - b^2)y = (a^2 + b^2 - c^2)z;$$

then (x, y, z) are the coordinates of a point on the same cubic curve.

This being so, it is required to find the geometrical relation of the two points to each other.

Solution by Professor Cayley.

1. On referring to Professor Wolstenholme's Solution of the original Question 3067 (*Reprint*, Vol. xiii., p. 70), it appears that the coordinates (x, y, z) of the point in question may be expressed in the more simple form

$$x : y : z = a(-a + b + c) : b(a - b + c) : c(a + b - c);$$

viz. the given relation between (a, b, c) being equivalent to

$$4abc + (-a + b + c)(a - b + c)(a + b - c) = 0,$$

we have

$$a^2 - (b - c)^2 = \frac{-4abc}{-a + b + c},$$

and thence

$$b^2 + c^2 - a^2 = 2bc \left(1 + \frac{2a}{-a + b + c} \right) = 2bc \left(\frac{a + b + c}{-a + b + c} \right);$$

and consequently

$$(b^2 + c^2 - a^2)x = \frac{2abc(a + b + c)x}{a(-a + b + c)},$$

whence the transformation in question.

2. Writing for greater symmetry (x, y, z) in place of (a, b, c) , and (x', y', z') in place of (x, y, z) , the coordinates (x, y, z) and (x', y', z') of the two points are connected by the relation

$$x' : y' : z' = x(-x + y + z) : y(x - y + z) : z(x + y - z),$$

and we thence at once deduce the converse relation

$$x : y : z = x'(-x' + y' + z') : y'(x' - y' + z') : z'(x' + y' - z').$$

Hence, writing

$$(-x + y + z, \quad x - y + z, \quad x + y - z) = (\xi, \eta, \zeta),$$

and similarly

$$(-x' + y' + z', \quad x' - y' + z', \quad x' + y' - z') = (\xi', \eta', \zeta'),$$

we have

$$x' : y' : z' = x\xi : y\eta : z\zeta, \quad x : y : z = x'\xi' : y'\eta' : z'\zeta',$$

and thence also $\xi\xi' = \eta\eta' = \zeta\zeta'$; so that, regarding $(\xi, \eta, \zeta), (\xi', \eta', \zeta')$ as the coordinates of the two points, we see that these are inverse points one of the other in regard to the triangle $\xi = 0, \eta = 0, \zeta = 0$.

To complete the solution, we must introduce these new coordinates into the equation of the cubic curve. Writing this under the form

$$8xyz + 2(-x + y + z)(x - y + z)(x + y - z) = 0,$$

and observing that

$$(2x, 2y, 2z) = (\eta + \zeta, \zeta + \xi, \xi + \eta),$$

the equation is

$$(\eta + \zeta)(\zeta + \xi)(\xi + \eta) + 2\xi\eta\zeta = 0;$$

viz. this is a cubic curve inverting into itself. And the two points in question are thus any two inverse points on this cubic curve.

3. In regard to the original form, that the point (x, y, z) defined by the equations

$$x(-a^2 + b^2 + c^2) = y(a^2 - b^2 + c^2) = z(a^2 + b^2 - c^2),$$

lies on the cubic curve

$$a^3 + b^3 + c^3 - (b + c)(c + a)(a + b) = 0,$$

Professor Sylvester proceeds as follows:—Writing

$$(x, y, z) = \{a^4 - (b^2 - c^2)^2, \quad b^4 - (c^2 - a^2)^2, \quad c^4 - (a^2 - b^2)^2\} = (A, B, C),$$

suppose; and

$$F(a, b, c) = a^3 + b^3 + c^3 - (b + c)(c + a)(a + b),$$

he observes that the truth of the theorem depends on the identity

$$F(A, B, C) + F(a, b, c)F(a, -b, c)F(a, b, -c)F(a, -b, -c) = 0,$$

and that, in order to prove the identity generally, it is sufficient to prove it for the three cases $a^2 = 0, a^2 = b^2 + c^2, a^2 = b^2$, which may be effected without difficulty.

4. But, for a general proof of the identity, write

$$\lambda = b^2 + c^2, \quad \mu = b^2 - c^2,$$

so that

$$A = a^4 - \mu^2, \quad B = (a^2 + \mu)(-a^2 + \lambda), \quad C = (-a^2 + \lambda)(a^2 - \mu),$$

whence

$$\begin{aligned} -F(A, B, C) &= -(a^4 - \mu^2)^3 + 2(a^2 - \lambda)^3(a^6 + 3a^2\mu^2) - 8a^2b^2c^2(a^4 - \mu^2)(a^2 - \lambda) \\ &= a^{12} - 6\lambda a^{10} + (6\lambda^2 + 9\mu^2 - 8b^2c^2)a^8 + \lambda(-2\lambda^3 - 18\mu^2 + 8b^2c^2)a^6 \\ &\quad + \mu^2(18\lambda^2 - 3\mu^2 + 8b^2c^2)a^4 + \lambda\mu^2(-6\lambda^2 - 8b^2c^2)a^2 + \mu^6. \end{aligned}$$

Moreover

$$F(a, b, c) = a\{a^2 - (b + c)^2\} - (b + c)\{a^2 - (b - c)^2\},$$

therefore

$$F(a, -b, -c) = a\{a^2 - (b + c)^2\} + (b + c)\{a^2 - (b - c)^2\};$$

whence

$$\begin{aligned} F(a, b, c)F(a, -b, -c) &= a^2\{a^2 - (b + c)^2\}^2 - (b + c)^2\{a^2 - (b - c)^2\}^2 \\ &= a^6 - 3\gamma^2a^4 + \gamma^2(\gamma^2 + 2\delta^2)a^2 - \gamma^2\delta^4, \end{aligned}$$

if $\gamma = b + c$, $\delta = b - c$. By changing the sign of c , we interchange γ and δ , and we thus have

$$F(a, b, -c)F(a, -b, c) = a^6 - 3\delta^2a^4 + \delta^2(2\gamma^2 + \delta^2)a^2 - \gamma^4\delta^2,$$

and the identity to be verified is thus

$$\begin{aligned} &\{a^6 - 3\gamma^2a^4 + \gamma^2(\gamma^2 + 2\delta^2)a^2 - \gamma^2\delta^4\} \\ &\quad \times \{a^6 - 3\delta^2a^4 + \delta^2(2\gamma^2 + \delta^2)a^2 - \gamma^4\delta^2\} \\ &= a^{12} - 6\lambda a^{10} + \dots + \mu^6, \quad \text{ut supra;} \end{aligned}$$

the values of λ, μ in terms of γ, δ are $\lambda = \frac{1}{2}(\gamma^2 + \delta^2)$, $\mu = \gamma\delta$; substituting these values on the right-hand side, the verification can be completed without difficulty.

[Vol. xxv., January to July, 1876, p. 82.]

4946. (Proposed by Professor Cayley.)—Show that the attraction of an indefinitely thin double convex lens on a point at the centre of one of its faces is equal to that of the infinite plate included between the tangent plane at the point and the parallel tangent plane of the other face of the lens.

[Vol. xxvi., July to December, 1876, pp. 41, 42.]

5020. (Proposed by W. S. B. Woolhouse, F.R.A.S.)—Let $1, \delta_1, \delta_2, \delta_3, \dots, \delta_n$ be the first differences of the coefficients of the expansion of the binomial $(1+x)^{2n}$ taken as far as the central or maximum coefficient; also let

$$\nu = \frac{1}{2}(n+1)n, \quad \nu' = \frac{1}{2}n(n-1), \quad \nu'' = \frac{1}{2}(n-1)(n-2), \quad \&c.;$$

then show that the algebraic function

$$x^\nu - \delta_1 x^{\nu'} + \delta_2 x^{\nu''} - \delta_3 x^{\nu'''} + \&c.$$

is divisible by $(x-1)^n$ without a remainder; and that the sum of the numerical coefficients of the quotient is equal to $1.3.5 \dots 2n-1$.

[See Solution to Question 1894, *Reprint*, vol. v., p. 113.]

Solution by Professor Cayley.

Mr Woolhouse's elegant theorem depends ultimately on the property of triangular numbers $\phi(n) = \frac{1}{2}(n^2 - n)$; then $\phi(n+1) = \phi(-n)$, so that, writing down the series of triangular numbers backwards and forwards,

$$\dots, 10, 6, 3, 1, 0, 0, 1, 3, 6, 10, \dots,$$

$$\dots, \quad, a, b, c, d, e, f, g, h, \quad \dots$$

we have, in fact, a continuous single series obtained by giving to n the different negative and positive integer values, zero included.

Thus a particular case is

$$(1+x)^6 - 5(1+x)^3 + 9(1+x) - 5 = 0 \pmod{x^3} = 1.3.5x^3 + \&c. x^4 + \dots,$$

where, on the left-hand side, the exponents are the triangular numbers $\phi(n+1)$, $n = 0$ to 3; and the coefficients, after the first, are the differences of the binomial coefficients of the power $2n$ (in the particular case, $n = 3$); viz. the binomial coefficients being

$$1, \quad 6, \quad 15, \quad 20, \quad 15, \quad 6, \quad 1,$$

the differences taken as far as they are positive are

$$5, \quad 9, \quad 5.$$

Expanding the several terms and writing down only the coefficients, we have a diagram

$$\begin{array}{ccccccc} & 1, & 6, & 15, & 20, & 15, & 6, & 1, \\ -5 & 1, & 3, & 3, & 1, & & & \\ +9 & 1, & 1, & & & & & \\ -5 & 1, & & & & & & \end{array}$$

The theorem in the particular case depends on the identities

$$\begin{array}{rcl} 1 - 5 + 9 - 5 & = & 0, \\ 6 - 15 + 9 & = & 0, \\ 5 - 15 & = & 0, \\ 20 - 5 & = & 1.3.5; \end{array}$$

or writing, as above, h, g, f, e , to denote the triangular numbers 6, 3, 1, 0, these may be replaced by

$$\begin{array}{rcl} h^0 & -5g^0 & +9f^0 - 5e^0 = 0, \\ h & -5g & +9f - 5e = 0, \\ \frac{1}{2}h(h-1) & -5 \cdot \frac{1}{2}g(g-1) & + \dots = 0, \\ \frac{1}{6}h(h-1)(h-2) & -5 \cdot \frac{1}{6}g(g-1)(g-2) & + \dots = 1.3.5; \end{array}$$

or, reducing each equation by those which precede it, these become

$$\begin{aligned}h^0 - 5g^0 + 9f^0 - 5e^0 &= 0, \\h^1 - 5g^1 + 9f^1 - 5e^1 &= 0, \\h^2 - 5g^2 + 9f^2 - 5e^2 &= 0, \\h^3 - 5g^3 + 9f^3 - 5e^3 &= 1.2.3.1.3.5.\end{aligned}$$

Consider any one of these, for instance the third; the function on the left-hand is

$$1h^2 - (6-1)g^2 + (15-6)f^2 - (20-15)e^2,$$

or, introducing the values b, c, d as above,

$$1h^2 - 6g^2 + 15f^2 - 20e^2 + 15d^2 - 6c^2 + 1b^2;$$

which is, in fact, $= 0$, if b, c, d, e, f, g, h are any successive triangular numbers; viz. this is an immediate consequence of the well-known theorem

$$\begin{aligned}1(\theta+6)^m - 6(\theta+5)^m + 15(\theta+4)^m - 20(\theta+3)^m + 15(\theta+2)^m - 6(\theta+1)^m + \theta^m \\= \Delta^6 \theta^m, \quad = 0 \text{ for any value of } m \text{ up to } m = 5, \\= 1.2.3.4.5.6 \text{ for } m = 6.\end{aligned}$$

We have thus all the equations except the last; and as regards the last equation, observe that the equation to be verified is

$$1\{\tfrac{1}{2}(\theta+6)(\theta+5)\}^3 - 6\{\tfrac{1}{2}(\theta+5)(\theta+4)\}^3 + \dots = 1.2.3.1.3.5,$$

viz. this may be replaced by

$$1(\theta+6)^6 - 6(\theta+5)^6 + \dots = 2^3.1.2.3.1.3.5 = 2.4.6.1.3.5 = 1.2.3.4.5.6,$$

which is right.

It is clear that the proof, although worked out on a particular case, is perfectly general; and Mr Woolhouse's theorem is thus proved.

[Vol. xxvi., pp. 77, 78.]

5079. (Proposed by Professor Cayley.)—Show that the curve

$$\begin{aligned}&\{(\beta - \gamma i)^2 - \delta^2\}^{\frac{1}{2}} \{(x - \beta i)^2 + y^2\}^{\frac{1}{2}} + q\{(\beta + \gamma i)^2 - \delta^2\}^{\frac{1}{2}} \{(x + \beta i)^2 + y^2\}^{\frac{1}{2}} \\&= \left\{ (1 - q^2) \frac{\beta}{\delta} \right\}^{\frac{1}{2}} \{\beta^2 - (\gamma - \delta i)^2\}^{\frac{1}{2}} \{(x - \gamma - \delta i)^2 + y^2\}^{\frac{1}{2}},\end{aligned}$$

where $i = (\sqrt{-1})$ as usual, is a real bicircular quartic having the axial foci

$$\beta i, \quad -\beta i, \quad \gamma + \delta i, \quad \gamma - \delta i.$$

Solution by the Proposer.

Consider the equation

$$(l + mi)^{\frac{1}{2}} \{(x - \beta i)^2 + y^2\}^{\frac{1}{2}} + q(l - mi)^{\frac{1}{2}} \{(x + \beta i)^2 + y^2\}^{\frac{1}{2}} = (\lambda + \mu i)^{\frac{1}{2}} \{(x - \gamma - \delta i)^2 + y^2\}^{\frac{1}{2}}.$$

This is

$$\begin{aligned} & (l + mi)\{x^2 + y^2 - \beta^2 - 2\beta xi\} + q^2(l - mi)\{x^2 + y^2 - \beta^2 + 2\beta xi\} \\ & - (\lambda + \mu i)\{x^2 + y^2 - \beta^2 + \gamma^2 - \delta^2 - 2\gamma x - 2(x - \gamma)\delta i\} \\ & + 2q(l^2 + m^2)^{\frac{1}{2}} \{(x^2 + y^2 - \beta^2)^2 + 4\beta^2 x^2\}^{\frac{1}{2}} = 0, \end{aligned}$$

where, putting the imaginary part equal to zero, we have

$$m(1 - q^2)(x^2 + y^2 - \beta^2) - 2l(1 - q^2)\beta x - \mu\{x^2 + y^2 - \beta^2 + (\beta^2 + \gamma^2 - \delta^2) - 2\gamma x\} + 2\lambda(x - \gamma)\delta = 0,$$

which will be true identically if

$$\begin{aligned} m(1 - q^2) - \mu &= 0, \\ -l(1 - q^2)\beta + \mu\gamma + \lambda\delta &= 0, \\ -\mu(\beta^2 + \gamma^2 - \delta^2) - 2\lambda\gamma\delta &= 0. \end{aligned}$$

The last gives

$$\lambda = \theta(\beta^2 + \gamma^2 - \delta^2), \quad \mu = -2\theta\gamma\delta, \quad \theta \text{ arbitrary};$$

and then

$$\begin{aligned} l(1 - q^2)\beta &= \theta\delta(\beta^2 + \gamma^2 - \delta^2 - 2\gamma^2) = \theta\delta(\beta^2 - \gamma^2 - \delta^2), \\ m(1 - q^2) &= -2\theta\delta\gamma; \end{aligned}$$

so that, putting

$$\theta\delta = (1 - q^2)\beta, \quad \text{or} \quad \theta = (1 - q^2)\frac{\beta}{\delta},$$

we have

$$\begin{aligned} l &= \beta^2 - \gamma^2 - \delta^2, \quad m = -2\beta\gamma, \\ \lambda &= (1 - q^2)\frac{\beta}{\delta}(\beta^2 + \gamma^2 - \delta^2), \quad \mu = (1 - q^2)\frac{\beta}{\delta}(-2\gamma\delta). \end{aligned}$$

Therefore

$$\begin{aligned} l \pm mi &= (\beta \mp \gamma i)^2 - \delta^2, \\ \lambda \pm \mu i &= (1 - q^2)\frac{\beta}{\delta} \{\beta^2 + (\gamma \mp \delta i)^2\}; \end{aligned}$$

and the equation is

$$\begin{aligned} & \{(\beta - \gamma i)^2 - \delta^2\}^{\frac{1}{2}} \{(x - \beta i)^2 + y^2\}^{\frac{1}{2}} + q\{(\beta + \gamma i)^2 - \delta^2\}^{\frac{1}{2}} \{(x + \beta i)^2 + y^2\}^{\frac{1}{2}} \\ & = \left\{ (1 - q^2)\frac{\beta}{\delta} \right\}^{\frac{1}{2}} \{\beta^2 - (\gamma - \delta i)^2\}^{\frac{1}{2}} \{(x - \gamma - \delta i)^2 + y^2\}^{\frac{1}{2}}, \end{aligned}$$

which is a real curve having the axial foci $+\beta i$, $-\beta i$; $\gamma + \delta i$, $\gamma - \delta i$; viz. $\gamma + \delta i$ being a focus, and the curve being real, it is clear that $\gamma - \delta i$ is also a focus.

[Vol. xxvii., January to June, 1877, p. 20.]

5130. (Proposed by Professor Cayley.)—Show that the envelope of a variable circle, having its centre on a given conic and cutting at right angles a given circle, is a bicircular quartic; which, when the given conic and the circle have double contact, becomes a pair of circles; and, by means of the last-mentioned particular case of the theorem, connect together the porisms arising out of the two problems—

- (i) Given two conics, to find a polygon of n sides inscribed in the one and circumscribed about the other.
- (ii) Given two circles, to find a closed series of n circles each touching the two circles and the two adjacent circles of the series.

[Vol. xxvii., pp. 81–83.]

5208. (Proposed by Professor Sylvester.)—Let the *magnitude* of any ramification signify the number of its branches, and let its partial magnitudes in respect to any node signify the magnitudes of the ramifications which come together at that node. If at any node the largest magnitude exceeds by k the sum of the other magnitudes, let the node be called superior by k , or be said to be of superiority k ; but if no magnitude exceeds the sum of the other magnitudes, let the node be called subequal. Then the theorem is, in any ramification, *either* there is one and only one subequal node; *or* else there are two and only two nodes each superior by unity, these two nodes being contiguous.

Solution by Professor Cayley.

The proof consists in showing that (1) there cannot be more than one subequal node; (2) there cannot be more than two nodes each superior by unity: and if there is one such node, then there is, contiguous to it, another such node; (3) starting from a node which is superior by more than unity, there is always a contiguous node which is either of smaller superiority, or else subequal; for, these theorems holding good, we can, by (3), always arrive at a node which is either subequal or else superior by unity; in the former case, by (1), the subequal node thus arrived at is unique; in the latter case, by (2), we have, contiguous to the node arrived at, a second node superior by unity; and we have thus a unique pair of nodes each superior by unity.

I will prove only (3), as it is easy to see that the like process applies to the proof of (1) and (2).

Let the whole magnitude be n ; and suppose at a node P which is superior by k , the largest magnitude is α , and that the other magnitudes are, say, β, γ, δ . We

have $\alpha = \beta + \gamma + \delta + k$; and since $n = \alpha + \beta + \gamma + \delta$, we have thence $n = 2\alpha - k$, or

$$\alpha = \frac{1}{2}(n + k), \quad \beta + \gamma + \delta = \frac{1}{2}(n - k) :$$

clearly k is even or odd, according as n is even or odd.

Suppose now that we pass from P , along the branch of magnitude α , to a contiguous node Q ; and let the magnitudes for Q be $\alpha', \beta', \gamma', \delta', \epsilon'$, where α' denotes the magnitude for the branch QP . We have $\alpha' = \beta + \gamma + \delta + 1$: for the ramification consists of the branch QP and of the ramifications of magnitudes β, γ, δ which meet in P . We have thus

$$\alpha' = \frac{1}{2}(n - k) + 1 = \frac{1}{2}n - \frac{1}{2}(k - 2);$$

and thence

$$\beta' + \gamma' + \delta' + \epsilon' = \frac{1}{2}n + \frac{1}{2}(k - 2).$$

Supposing here that k is greater than 1, viz. that it is = or > 2 , $k - 2$ is 0 or positive; and if α' be the greatest magnitude belonging to the node Q , this is a subequal node. But it may be that α' is not the greatest magnitude; supposing then that the greatest magnitude is β' , we have

$$\beta' = \frac{1}{2}n + \frac{1}{2}(k - 2) - \gamma' - \delta' - \epsilon',$$

$$\alpha' + \gamma' + \delta' + \epsilon' = \frac{1}{2}n - \frac{1}{2}(k - 2) + \gamma' + \delta' + \epsilon',$$

and thence

$$\beta' - (\alpha' + \gamma' + \delta' + \epsilon') = k - 2 - 2(\gamma' + \delta' + \epsilon');$$

viz. either the node is subequal, or else, being superior, the superiority is at most $= k - 2$; that is, if from the node P , of superiority = or > 2 , we pass along the branch of greatest magnitude to the contiguous node Q , this is either subequal, or else of superiority less than that of P ; which is the foregoing proposition (3).

The subequal node, and the two nodes of superiority 1, in the cases where they respectively exist, may be termed the centre and the bicentre respectively; and the theorem thus is, every ramification has either a centre or else a bicentre. But the centre and the bicentre here considered, due (as remarked by Professor Sylvester) to M. Camille Jordan, and which may for distinction be termed the centre and the bicentre of *magnitude*, are quite distinct from the centre and the bicentre discovered by Professor Sylvester, and considered in my researches upon trees, *British Association Report*, 1875, [610]. These last may for distinction be termed the centre and the bicentre of *distance*: viz. we here consider, not the magnitude, but the length of a ramification, such length being measured by the number of branches to be travelled over in order to reach the most distant terminal node. The ramification has either a centre or else a bicentre of distance: viz. the centre is a node such that, for two or more of the ramifications which proceed from it, the lengths are equal and superior to those of the other ramifications, if any; the bicentre a pair of contiguous nodes such that, disregarding the branch which unites the two nodes, there are from the two nodes respectively (one at least from each of them) two or more ramifications the lengths of which are equal to each other and superior to those of the other ramifications, if any.

It is very noticeable how close the agreement is between the proofs for the existence of the two kinds of centre or bicentre respectively. Say, first as regards distance, if at any node the length of the longest branch exceeds by k the length of the next longest branch or branches, then the node is superior by k , or is of the superiority k ; but, if there are two or more longest branches, then the node is subequal. And say next, in regard to magnitude, if at any node the largest magnitude exceeds by k the sum of all the other magnitudes, the node is superior by k , or has a superiority k ; but if the largest magnitude does not exceed the sum of the other magnitudes, then the node is subequal. Then, whether we attend to distance or to magnitude, the three propositions hold good: (1) there cannot be more than one subequal node; (2) there cannot be more than two nodes each superior by unity: and if there is one such node, there is contiguous to it another such node; (3) starting from a node which is superior by more than unity, there is always a contiguous node which is of smaller superiority or else subequal; whence, as in the solution just referred to, there is always, as regards distance, a centre or a bicentre; and there is always, as regards magnitude, a centre or a bicentre.

[Vol. xxvii., pp. 89, 90.]

On Mr Artemas Martin's First Question in Probabilities. By Professor Cayley.

The question was, “ A says that B says that a certain event took place: required the probability that the event did take place, p_1 and p_2 being A 's and B 's respective probabilities of speaking the truth.”

The solutions, referred to or given on pp. 77–79 of volume xxvii. of the *Reprint*, give the following values for the probability in question:—

Todhunter's <i>Algebra</i>	$p_1p_2 + (1 - p_1)(1 - p_2).$
Artemas Martin	$p_1[p_1p_2 + (1 - p_1)(1 - p_2)].$
American Mathematicians and Woolhouse	$p_1p_2.$

It seems to me that the true answer cannot be expressed in terms of only p_1 and p_2 , but that it involves two other constants β and k ; and my value is—

$$\text{Cayley} p_1p_2 + \beta(1 - p_1)(1 - p_2) + k(1 - \beta)(1 - p_1).$$

In obtaining this I introduce, but I think of necessity, elements which Mr Woolhouse calls extraneous and imperfect.

B told A that the event happened, or he did not tell A this; the only evidence is A 's statement that B told him that the event happened; and the chances are p_1 and $1 - p_1$. But, in the latter case, either B told A that the event did not happen, or he did not tell him at all; the chances (on the supposition of the incorrectness of A 's statement) are β and $1 - \beta$; and the chances of the three cases